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RESULTS

Results

01 · LOGISTIC REGRESSION

predict a yes/no outcome

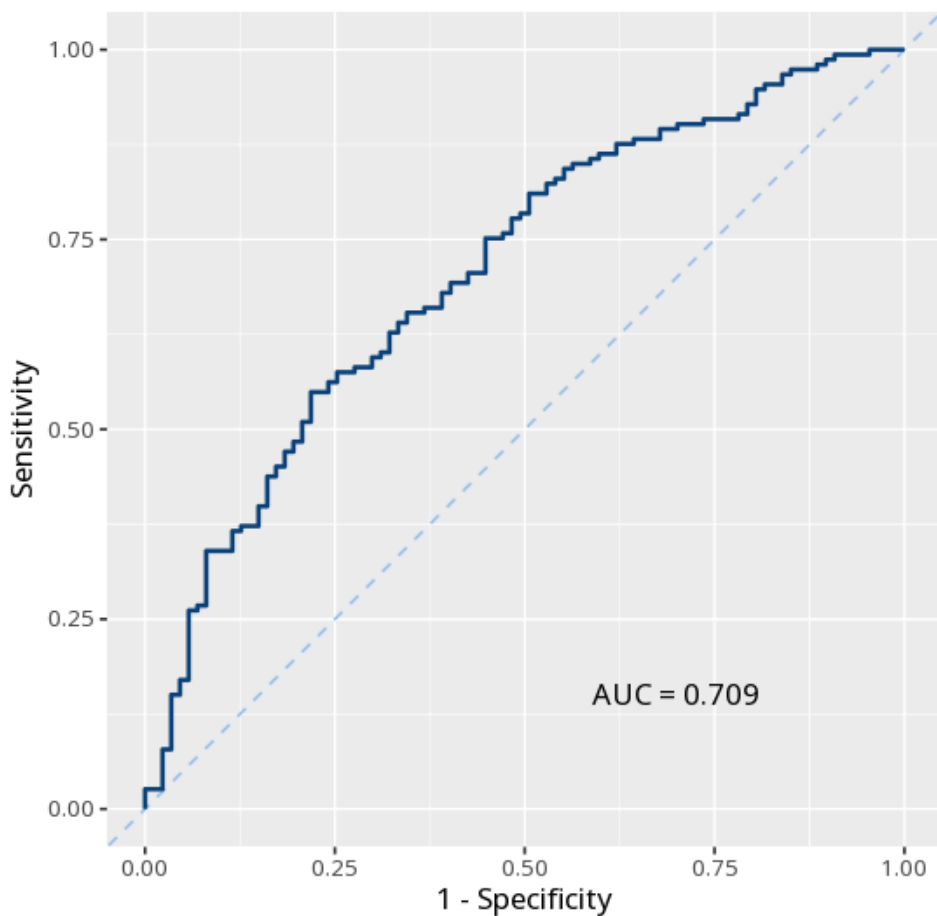
Table 1. Coefficients

	Log-odds (B)	Odds ratio (OR)
(Intercept)	−2.69 (0.96)	0.07 [< 0.01, 0.43]
pretest_score	0.04 (0.02)	1.04 [1.01, 1.08]
study_hours_per_week	0.19 (0.06)	1.21 [1.08, 1.37]
teaching_method: Flipped	0.96 (0.37)	2.61 [1.27, 5.52]
teaching_method: Lecture	−0.50 (0.33)	0.60 [0.31, 1.16]
Num.Obs.	240	
Nagelkerke R ²	0.17	
Omnibus χ^2	31.41 (p < .001)	
Log.Lik.	−141.46	
AIC	292.91	
BIC	310.31	
Accuracy	69.2%	
Sensitivity	86.9%	
Specificity	37.9%	
AUC [95% CI]	.71 [.64, .78]	

Table 2. Classification

Predicted \ Observed	No	Yes	% correct
No	33	20	62.3%
Yes	54	133	71.1%

Figure. Classifier performance



Understanding the numbers

Odds ratio (OR). How the odds of the outcome multiply per one-unit increase in the predictor (>1 raises the odds, <1 lowers them). Here, $OR = 1.04$, so a one-unit increase in the predictor multiplies the odds by this factor.

Log-odds (B). The raw coefficient on the log-odds scale that the odds ratio is exponentiated from. Here, $B = 0.04$ on the log-odds scale.

Omnibus χ^2 . Tests the model as a whole against a model with no predictors. Here, the omnibus $\chi^2 = 31.41$ ($p < .001$).

Nagelkerke R^2 . A pseudo- R^2 approximating the share of variance explained (0 to 1, higher is more). Here, Nagelkerke $R^2 = 0.17$.

Log.Lik.. The log-likelihood: how probable the observed outcomes are under the fitted model; it underlies AIC and BIC. Here, $\text{Log.Lik.} = -141.46$.

AIC. A fit measure that penalizes model complexity, useful for comparing models fit to the same data (lower is better). Here, $AIC = 292.91$.

BIC. Like AIC but penalizes complexity more heavily as sample size grows (lower is better). Here, $BIC = 310.31$.

N. The number of observations used to fit the model. Here, $N = 240$.

Predicted \ Observed (first category). The diagonal (correct) cell for the first outcome category: how many cases the model predicted into it that were actually observed there. Here, 33 cases were predicted and observed in the first category.

Predicted \ Observed (second category). The diagonal (correct) cell for the second outcome category: how many cases the model predicted into it that were actually observed there. Here, 133 cases were predicted and observed in the second category.

% correct. The overall share of cases the model classified correctly - model fit and the ROC/AUC show how well it classifies overall. Here, the model classified 69.2% of cases correctly.

Accuracy. The overall share of cases classified correctly (same number as % correct, reported as a named scalar here). Here, accuracy = 69.2%.

Sensitivity. The share of ACTUAL events the model correctly predicted as events (the true positive rate). Here, sensitivity = 86.9%.

Specificity. The share of ACTUAL non-events the model correctly predicted as non-events (the true negative rate). Here, specificity = 37.9%.

AUC [95% CI]. The area under the ROC curve, with its confidence interval (pROC's DeLong method) - 0.5 is chance, 1.0 is perfect discrimination. Here, AUC = .71 [.64, .78].

How to read this test

Each predictor's odds ratio tells how the odds of the outcome change per unit (>1 increases, <1 decreases); a term carries weight when its OR confidence interval excludes 1, and the inline omnibus χ^2 (with its p) tests the model as a whole. Model fit and the ROC/AUC show how well it classifies overall, alongside accuracy (overall % correct), sensitivity (% of actual events correctly predicted), and specificity (% of actual non-events correctly predicted). Your significance threshold (α) is 0.05.

APA template: Predictor pretest_score was associated with the outcome, OR=1.04, 95% CI [1.01, 1.08], p = .025 (AUC=.71).

Statistical basis: Logistic regression coefficients (log-odds / odds ratios) (R Core Team (2026). R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna.); Nagelkerke pseudo- R^2 (Lüdtke D, Ben-Shachar MS, Patil I, Waggoner P, Makowski D (2021). "performance: An R Package for Assessment, Comparison and Testing of Statistical Models." Journal of Open Source Software, 6(60), 3139.); ROC curve / AUC (Robin X, Turck N, Hainard A, et al. (2011). "pROC: an open-source package for R and S+ to analyze and compare ROC curves." BMC Bioinformatics, 12, 77.). Full references in the exported CITATIONS.txt.

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